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BRIEF REVIEW

The Relationship Between Isometric and Dynamic Strength Following Resistance Training: A Systematic Review, Meta-Analysis, and Level of Agreement

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Background: Maximal lower-body strength can be assessed both dynamically and isometrically; however, the relationship between the changes in these 2 forms of strength following resistance training is not well understood. ***Purpose:*** To systematically review and analyze the effects of resistance training on changes in maximal dynamic (1-repetition-maximum back squat, deadlift, and power clean) and position-matched isometric strength (isometric midthigh pull and the isometric squat). In addition, individual-level data were used to quantify the agreement and relationship between changes in dynamic and isometric strength. ***Methods:*** Databases were systematically searched to identify eligible articles, and meta-analysis procedures were performed on the extracted data. The raw results from 4 studies were acquired, enabling bias and absolute reliability measures to be calculated using Bland–Altman test of agreement. ***Results:*** Eleven studies met the inclusion criteria, which resulted in 29 isometric–dynamic change comparisons. The overall pooled effect was 0.13 in favor of dynamic testing; however, the prediction interval ranged from $g = -0.49$ to 0.75. There was no evidence of bias ($P = .825$) between isometric and dynamic tests; however, the reliability coefficient was estimated to be 16%, and the coefficient of variation (%) was 109.27. ***Conclusions:*** As a range of future effects can be expected when comparing isometric to dynamic strength changes following resistance training, and limited proportionality exists between changes in these 2 strength qualities, there is strong evidence that isometric and dynamic strength represent separate neuromuscular domains. These findings can be used to inform strength-assessment models in athlete populations.

Keywords: testing, assessment, validity, adaptations

The assessment of specific fitness qualities is a key step in the training process, as it can be used to quantify the effects of training and can be associated with sport performance outcomes.[1] Furthermore, athlete assessments can inform decisions regarding training prescription and the management of fitness and fatigue across time. For example, regular monitoring with a countermovement jump[2] or of barbell velocity during a primary training lift[3] can help identify acute changes in fatigue state, which practitioners can consider when prescribing training loads. An important consideration of performance testing is to establish whether tests are similar or distinct from each other.[4] A test can be considered distinct if it shares limited commonality or if the responses to training or fatigue are different over time when compared with other tests. Conversely, similar tests contain considerable overlapping information when compared cross-sectionally or longitudinally.[5–7] Assessment systems should be specific enough to isolate and track independent attributes while also minimizing redundant information.[8]

In many cases, the most important physical quality that requires assessment in sport is maximal strength.[9] This quality refers to the force-generating capacity of the athlete against an external resistance.[10] The gold standard of maximal dynamic strength assessment is a 1-repetition maximum (1RM) test, with variations involving multiple, but few, repetitions also considered acceptable (eg, 3RM or 5RM).[11] Maximal dynamic strength consistently distinguishes higher from lower level athletes within a range of sports and is strongly associated with other physical qualities and key performance indicators within competition.[9] The assessment of maximal dynamic strength is highly reliable across populations,[12–14] requires minimal equipment, can be conducted with large training groups, and is commonly used to prescribe intensity for upcoming training cycles. For these reasons, it is the key strength assessment method used by practitioners in sport.

Maximal strength can also be assessed isometrically, when force is applied maximally against an immovable resistance with no time constraints to the duration of force application.[15] Strength qualities are often presented along a continuum from high-velocity, low-force expressions to high-force, low-velocity expressions.[16–18] As isometric strength occurs at the extreme end of this continuum at zero velocity, it is often referred to as "pure" strength[19] and is therefore relevant in practice and research as it may represent an individual's ultimate strength capacity. While historically isometric measures of maximal force have held limited associations to dynamic performance,[15] when completed in a position that replicates a common athletic action, such as an isometric squat or isometric mid-thigh pull (IMTP), stronger relationships have been reported.[20] These isometric assessments of maximal strength have become commonplace in strength assessment models as they are often simple to administer, induce less fatigue than traditional

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dynamic assessments of strength, and are considered by some to carry a reduced injury concern due to their isometric demands.[21] However, the isometric squat and IMTP require specialized testing equipment and are influenced by joint angles in the testing position.[22] Nonetheless, these tests are an appealing and popular option for athlete strength assessment.

Both dynamic and isometric maximal strength assessments are reliable and are feasible to administer in most training environments.[14,21,23] However, while a high commonality exists between position-matched maximal dynamic and isometric strength in competitive weightlifters,[24,25] in all other athlete populations, they generally demonstrate <50% shared variance[20] and can therefore be considered independent forms of strength that are more dissimilar than similar.[4,5] It is unclear how this relationship tracks across time, as an isolated association does not necessarily imply a similar response to a training stimulus. Such information can help practitioners to ascertain whether a change in an isometric strength test is an indicator of a change in dynamic strength (which is typically the training target), or whether they are distinct neuromuscular domains that need to be assessed and trained independently.

Concerns have been raised over the sensitivity of isometric strength assessments to long-term training[15]; however, this was based on mostly single joint or supported multijoint tests that often demonstrated limited association to dynamic performance. It is possible that upright, unsupported, multijoint isometric assessments may demonstrate different results. Furthermore, recent analysis has indicated limited concordance and agreement between cross-sectional assessments of isometric and dynamic strength.[26] It is therefore important to understand how well changes in isometric strength transfer to changes in dynamic strength following resistance training, so practitioners can make evidence-based training and testing decisions.

The aim of this study was to systematically review and analyze the effects of resistance training on changes in maximal dynamic and isometric strength in healthy adults. Furthermore, where available, individual data were used to quantify the agreement and relationship between changes in dynamic and isometric strength. Finally, subgroup analysis was conducted to determine whether the type of isometric test, method of training, intervention duration, or participant characteristics were influential factors on the effect.

Methods

Search Strategy

In accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses framework for systematic reviews, the databases Web of Science, SPORTDiscus, MEDLINE, and PubMed were systematically searched in April 2022 to identify eligible English language peer-reviewed original research articles. All terms and key words can be found in Table 1. Reference list of the selected studies were manually searched for eligible articles.

Selection Criteria

Search results were imported into EndNote (X9, Thomson Reuters) for removal of duplicates and further sorting. Title and abstract screening for suitability was independently undertaken by James and Weakley, followed by full-text assessment of the remaining articles. Studies were deemed eligible if (1) they were a resistance training study that included pretest and posttest results for a 1RM, 3RM, or 5RM free weight squat (back or front), power clean, or deadlift, in addition to peak force derived from an IMTP or an isometric squat; (2) the training duration was a minimum of 5 weeks; (3) participants were apparently healthy adults or athletes; (4) they were published in a peer-reviewed journal before March 2022; and (5) they were full text in English language. Those that did not meet the inclusion criteria were excluded. Data pertaining to the authors, isometric and dynamic test used, study duration, participant demographics and group designation, and training type were extracted from each study. Additionally, changes in physical performance within each relevant test were extracted and used for analysis. Means, SDs, and sample size were required for eligibility. Authors from all of the eligible studies were contacted for study data at the individual level to determine the level of agreement between the isometric and dynamic changes.

Assessment of Reporting Quality

Study quality was assessed by the scale first employed by Brughelli et al[27] for strength and conditioning-based training interventions,[28,29] as opposed to health care research and interventions. The tool includes a combination of items from the Delphi, Cochrane, and PEDro (Physiotherapy Evidence Database) instruments and assesses the following 10 factors: (1) inclusion criteria were clearly stated, (2) subjects were randomly allocated to groups, (3) intervention was clearly defined, (4) groups were tested for similarity at baseline, (5) use of a control group, (6) outcome variables were clearly defined, (7) assessments were practically useful, (8) duration of intervention practically useful, (9) between-group statistical analysis appropriate, and (10) point measures of variability.

Each domain can achieve a maximum of 2 points for a "yes" response, while 1 and 0 represent "maybe" and "no," respectively. The scores for each of the domains are then summed to provide a total study quality score ranging from 0 to 20.

Statistical Analyses

Meta-analysis was performed using the "metafor"[30] and "clubSandwich" packages in the R programming language. To pool effect sizes between isometric and dynamic tests, we computed standardized mean differences (SMDs) for each study. Uncertainty was expressed using 95% confidence intervals (CIs), calculated based on a *t* distribution. Prediction intervals (PI) were computed alongside the estimates, via a random-effects model, to convey the

Table 1 Database Search Terms and Keywords

Search category 1	Search category 2	Search category 3	Search category 4
"Adaptation" OR "adaptations" OR "Changes" OR "Training" OR "Longitudinal" OR "Intervention" OR "Development"	"Squat" OR "Deadlift" OR "Power clean" OR "Power-clean"	"Repetition maximum" OR "repetition - maximum" OR "RM" OR "maximum" OR "strength"	"Isometric squat" OR "Isometric midthigh pull" OR "Isometric mid-thigh pull" OR "Isometric mid thigh pull"

Note: All 4 categories combined using "AND."

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likely range of the true change in maximal strength in similar future studies. Unlike CIs that estimate the range of the true effect size, PIs account for both within-study sampling error and between-study heterogeneity. The borders of the PI are determined by considering the estimated effect size, the standard error of the effect size, and the desired level of confidence. Including PIs are important as they show a wider range of expected treatment effects compared with CIs and, thus, may lead to different conclusions. From an applied perspective, they allow practitioners to determine what is expected to occur in future settings.[31] Between-studies heterogeneity was estimated with Cochran Q and Higgins and Thompson I^2 statistics.

An initial model (intercept-only), using restricted maximum likelihood, was constructed to serve as the baseline model. To examine the effect of programming variables on maximal strength, study factors were added to the baseline model as fixed effects. These moderator variables included study subject (strong, recreational, and collegiate); training (heavy resistance training, combined heavy-high velocity training, and other training); isometric test (mid-thigh pull and squat); and duration (less than 10 wk and greater than 10 wk).

Training type was classified as "Heavy resistance training" if >66% of resistance training activities was at 75% 1RM or above, "Combined heavy-high velocity training" if training consisted of both heavy and ballistic/plyometric training, or "Other training" for remaining resistance training structures. Subjects were classified as strong if their 1RM back squat or deadlift was >1.90 × body mass, or 1RM power clean was >1.5 × body mass,[32,33] otherwise they were classified as collegiate if they were university or college athletes or recreational for the remaining participant groups. Training duration was stratified as <10 weeks or ≥10 weeks, while the isometric test type was classified as either the IMTP or isometric squat.

The effects of each moderator were estimated (along with 95% CIs and PIs, where appropriate), with all other factors held constant. Cohen's d effect sizes were used to describe the magnitude of the observed pairwise differences in standardized units and interpreted with the following descriptors: trivial (<0.2), small (0.2–0.59), moderate (0.6–1.19), large (1.2–1.99), and very large (2.0–4.0).[34]

Finally, we were able to collate the raw results from 4 studies and conduct a validity analysis of the participant's isometric and dynamic test results, specifically the change between pretest and posttest. To evaluate the variability of the measurements, we calculated the coefficient of variation (CV) for each variable of interest. The CV was obtained by dividing the standard deviation of the measurements by their mean and multiplying by 100. A lower CV indicates lower relative variability and greater precision in the measurements. To assess the reliability of the measurements, we computed the standard error of measurement (SEM). The SEM provides an estimate of the typical amount of measurement error associated with individual test scores. It was calculated by multiplying the standard deviation of the measurements by the square root of one minus the reliability coefficient. A smaller SEM value indicates higher reliability and greater precision in the measurements. To evaluate the agreement between 2 measurement methods, we performed a Bland–Altman analysis. This involved plotting the differences between the measurements obtained from method A and method B against their average. We then assessed the presence of any systematic bias or limits of agreement.

Relative reliability was examined with the concordance correlation coefficient (CCC). The CCC is a statistical measure used to assess the agreement or concordance between 2 sets of measurements. It takes into account both the precision (closeness of values) and accuracy (how well the measurements align with a reference value) of the measurements. The imprecision of the estimates are denoted with 95% CI. To ensure that the model estimates are robust, a sensitivity analysis was conducted using linear mixed models, with fixed effects for time (pre–post) and random effects for subject ID.

Results

Search Results

Following the removal of duplicates, 195 studies remained. Following title and abstract screening, 28 articles were brought forward for full-text screening. Once full-text assessment was completed, 11 articles remained and were included in the meta-analysis and review. Figure 1 describes the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) process applied to this review. The 11 studies resulted in 29 isometric–dynamic change comparisons stemming from multiple groups or multiple isometric or dynamic tests within a given study. In total, 229 participants were investigated (competitive athletes: n = 126; recreational or resistance-trained individuals: n = 103). Data sets from 4 studies were provided to enable analysis at the individual level (N = 94; competitive athletes: n = 61; recreational or resistance-trained individuals: n = 33).

Outcome of Assessment of Reporting Quality

Study quality assessment scores ranged from 15 to 20 on the 20-point scale with a mean and SD of 17.2 (1.4), indicating a high methodological quality of included articles based on this scale. Most common criteria not met was "use of a control group" (met in 3/12 studies), as it is typically not feasible to use control groups in athlete populations, and "inclusion criteria clearly stated" (fully met in 8/12 studies).

Study Characteristics

The IMTP featured across 15 comparisons in 5 studies, while the isometric squat was present in 14 comparisons within 6 studies. The 1RM back squat was featured in 9 studies, each containing 2 groups, representing 18 comparisons. Three studies, each with 2 groups, contained the 1RM deadlift, resulting in 6 comparisons. The 1RM power clean occurred in 2 studies, with one containing 2 groups and the other with 3 groups, resulting in 5 comparisons. There were no studies in the final analysis that contained multiple RM testing or the front squat. A summary of studies included in the analysis is presented in Table 2.

The 4 studies for which individual values were obtained (and therefore brought forward for Bland–Altman tests) contained 9 study effects. The IMTP and isometric squat featured on 5 and 4 occasions, respectively. With regard to the dynamic strength tests, the 1RM back squat occurred 4 times, while the 1RM power clean was included on 5 occasions.

Meta-Analysis

The between-study heterogeneity variance for the baseline model was estimated at $\tau^2 = .09$ (95% CI, .03 to .20), with an I^2 value of 62.7% (95% CI, 44.3% to 75.0%). The overall pooled effect comparing the magnitude of change in isometric and dynamic strength was estimated to be 0.13 in favor of dynamic testing; however, the PI ranged from $g = -0.49$ to 0.75, indicating

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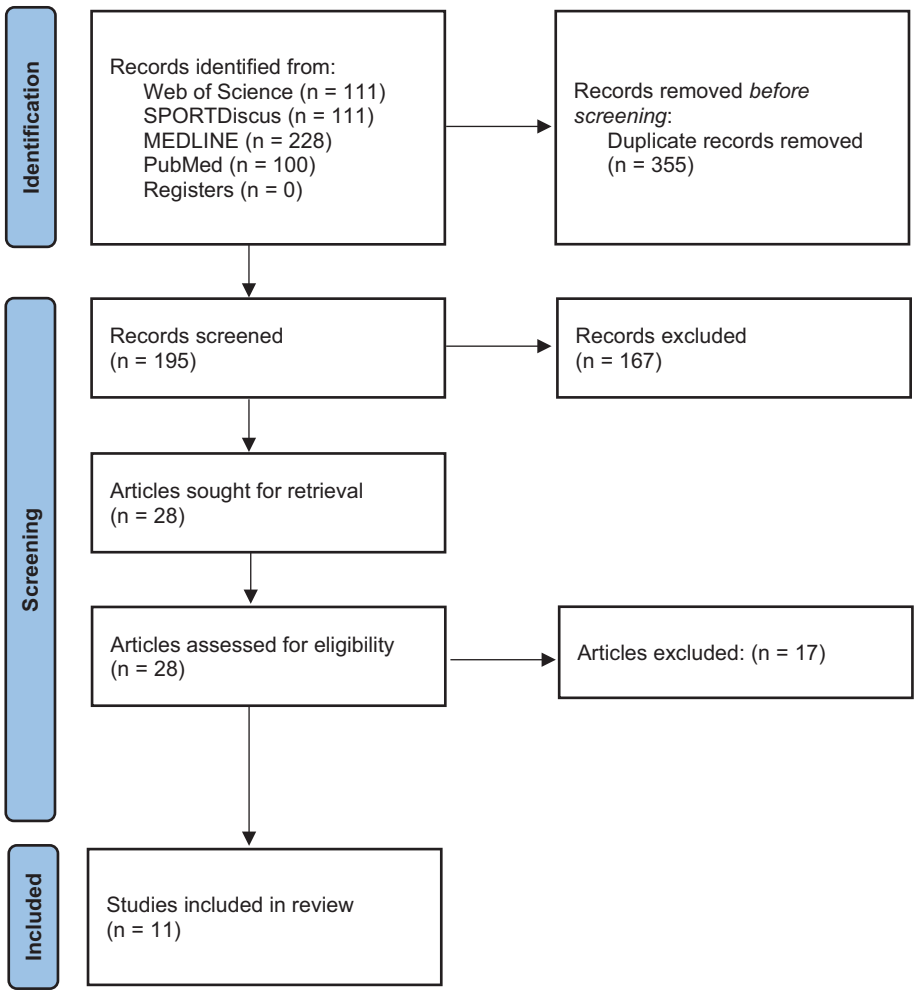

Figure 1 — PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flow diagram describing the selection processes for eligible articles.

uncertainty in the magnitude of change between isometric and dynamic tests for future studies (Figure 2).

Subgroup analyses (Table 3) indicated that the pooled effects are significant across several factors. These include heavy resistance training (SMD = 0.27; 95% PI, 0.10 to 0.45); duration less than 10 weeks (SMD = 0.15; 95% PI, 0.05 to 0.24); and recreational subjects (SMD = 0.27; 95% PI, 0.14 to 0.40). Figures for the subgroup analysis can be found in Supplementary Materials S1, S2, S3, and S4 (available online).

Bland–Altman Analysis

There was no evidence of bias ($P = .825$) between isometric and dynamic tests; however, the reliability coefficient (as assessed by concordance correlation) was estimated to be only 16%, indicating poor agreement between measurements on the same subject, with 95% limits of agreement (from Bland–Altman tests) estimated as −27.03 to 27.66. Based on a reliability analysis, the SEM was 9.82, with a large CV (%) of 109.27 and a poor CCC of .16.

The results of the sensitivity analysis (conducted via a linear mixed model) also found no evidence of bias between the 2 tests ($F_{1,93} = 0.05$, $P = .0825$), with a mean difference of 0.32 units between isometric and dynamic testing. The intraclass correlation coefficient was estimated to be equivalent to the CCC (.16) from the initial test. This demonstrates that the model estimates from the Bland–Altman test were robust and consistent across different methods.

Discussion

The primary aim of this investigation was to systematically review the evidence and analyze the effect of, and association between, training-induced changes in maximal dynamic strength with respect to changes in maximal isometric strength. In addition, this study sought to quantify the agreement and relationship between changes in the 2 forms of strength. The final aim was to examine the effect at several subgroup levels, including training type, test type, training duration, and participant characteristics. Eleven studies were included in the final review and analysis, with the results of the meta-analysis revealing that following resistance training there are trivial differences yet wide PIs when changes in isometric and dynamic strength are compared. This indicates that the magnitude of change in these different forms of strength test can be expected to vary dramatically following future resistance training interventions. A similar pattern was generally seen across most subgroups. Furthermore, the very small CI, wide limits of agreement, and absence of consistent bias revealed by the Bland–Altman analysis

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Table 2 Characteristics of Resistance-Training-Intervention Studies Containing Both Isometric and Dynamic Maximal Strength Assessment of the Lower Body

Study authors	Isometric test	Dynamic test	Duration, wk	Group 1 demographics (body mass)	Group 2 demographics (body mass)	Group 1 training type	Group 2 training type
Banaszek et al[35]	Midthigh pull	Back squat	8	7 recreational CrossFit males and females (83.9 [18.9])	8 recreational CrossFit males and females (78.4 [11.6])	CrossFit training (whey supplementation)	CrossFit training (pea supplementation)
Banaszek et al[35]	Midthigh pull	Deadlift	8	7 recreational CrossFit males and females (83.9 [18.9])	8 recreational CrossFit males and females (78.4 [11.6])	CrossFit training (whey supplementation)	CrossFit training (pea supplementation)
Bartolomei et al[36]	Squat	Back squat	10	10 experienced, resistance-trained men (78.7 [11.3])	11 experienced, resistance-trained men (79.2 [9.5])	Heavy resistance training (full body)	Heavy resistance training (split body)
Bazyler et al[37]	Squat	Back squat	7	9 recreationally trained males (84.9 [10.9])	8 recreationally trained males (84.6 [8.4] kg)	Heavy resistance training (full range of motion)	Heavy resistance training (partial range of motion)
Comfort et al[38]	Midthigh pull	Power clean	10	16 collegiate athletes and professional youth soccer players (71.14 [11.79])	18 collegiate athletes and professional youth soccer players (66.43 [10.13])	Combined heavy-high velocity training (catch phase)	Combined heavy-high velocity training (pull phase only)
Cormie et al[39]	Squat	Back squat	12	10 recreationally trained males (81.6 [18.8])	8 recreationally trained males (79.6 [15.4])	High velocity training	Combined heavy-high velocity training
Cormie et al[40]	Squat	Back squat	10	8 recreationally trained males (82.2 [13.7])	8 recreationally trained males (79.9 [14.5])	Heavy resistance training	High velocity training
James et al[32]	Squat	Back squat	10	8 strong males (76.82 [6.27])	8 recreationally active males (82.03 [14.7])	Combined heavy-high velocity training (stronger participants)	Combined heavy-high velocity training (weaker participants)
Painter et al[41]	Midthigh pull	Back squat	11	14 collegiate track athletes (86.1 [30.9])	12 collegiate track athletes (80.7 [18.1])	Combined heavy-high velocity training (block periodization)	Combined heavy-high velocity training (DUP)
Suchomel et al[42]	Midthigh pull	Power clean	10	9 collegiate males athletes (85.8 [13.4])	9 collegiate male athletes (84.3 [17.3])	Combined heavy-high velocity training (catch phase)	Combined heavy-high velocity training (pull phase only)
Suchomel et al[42]	Midthigh pull	Power clean	10	9 collegiate males athletes (83.0 [13.6])		Combined heavy-high velocity training (pull phase only + overload)	
Toohey et al[43]	Midthigh pull	Back squat	10	10 collegiate female athletes (70.0 [8.4])	13 collegiate female athletes (66.6 [5.1])	General resistance training (dietary supplement)	General resistance training (no supplement)
Toohey et al[43]	Midthigh pull	Deadlift	10	10 collegiate female athletes (70.0 [8.4])	13 collegiate female athletes (66.6 [5.1])	General resistance training (dietary supplement)	General resistance training (no supplement)
Travis et al[44]	Squat	Back squat	6	8 male and female powerlifters (88.6 [19.0])	8 male and female powerlifters (91.1 [24.9])	Heavy resistance training (step taper)	Heavy resistance training (exponential taper)
Travis et al[44]	Squat	Deadlift	6	8 male and female powerlifters (88.6[19.0])	8 male and female powerlifters (91.1 [24.9])	Heavy resistance training (step taper)	Heavy resistance training (exponential taper)

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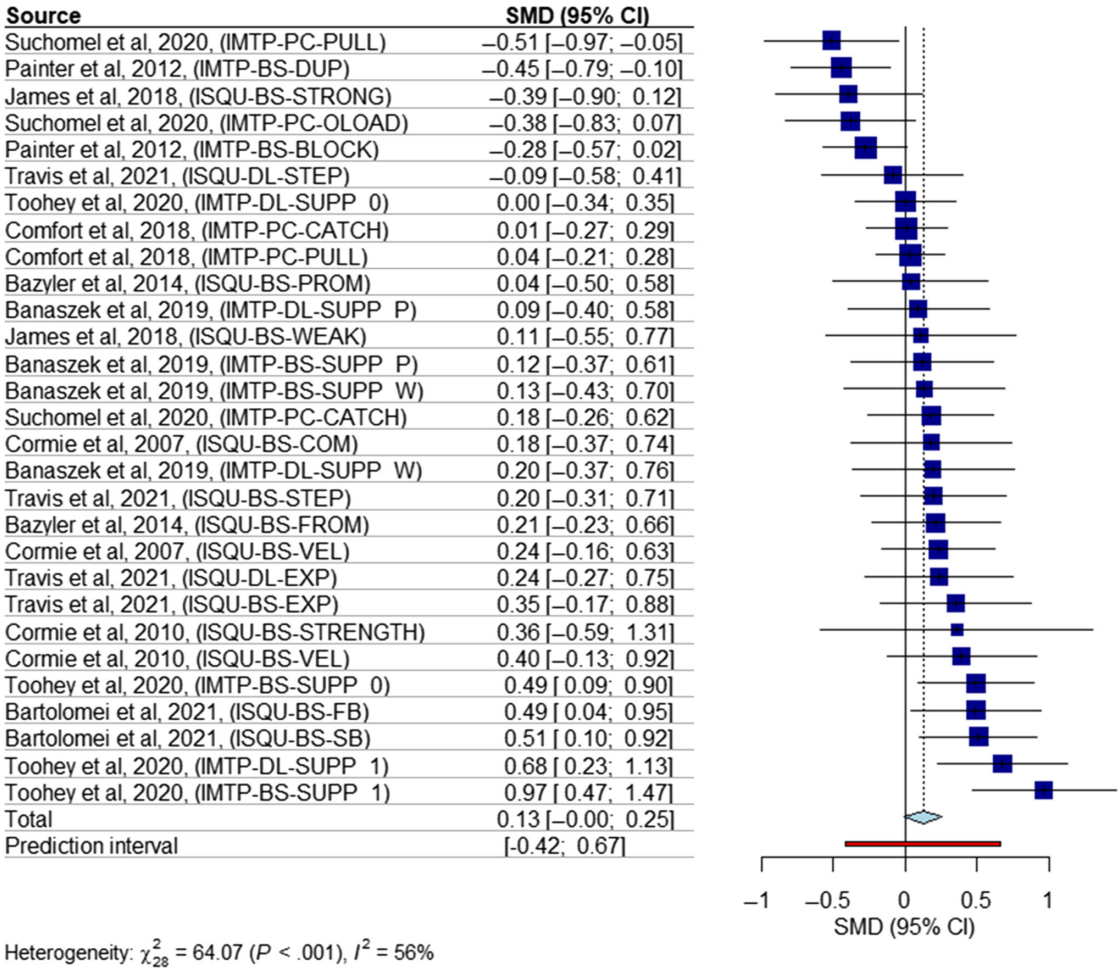

Figure 2 — Forest plot of the combined studies for the meta-analysis. SMD indicates standardized mean difference.

Table 3 Meta-Analyses With Subgroups

Subgroup	*k*	SMD	95% CI	95% PI	τ^2	*Q*	I^2 (%)
Training							
Heavy resistance training	9	0.27	0.10 to 0.44	0.10 to 0.45	.00	6.04	0.0
Combined heavy-high velocity	10	–0.18	–0.37 to 0.00	–0.55 to 0.18	.02	14.04	35.9
Other	10	0.38	0.13 to 0.62	–0.21 to 0.96	.05	17.26	47.8
Iso test							
Midthigh pull	15	0.06	–0.17 to 0.30	0.05 to 0.24	.13	52.66	73.4
Squat	14	0.23	0.07 to 0.40	–0.71 to 0.98	.01	14.88	12.6
Duration							
<10 wk	10	0.15	0.05 to 0.24	0.03 to 0.28	.00	2.36	0.0
10+ wk	19	0.14	–0.08 to 0.35	–0.56 to 0.91	.15	71.90	75.0
Subject							
Recreational	13	0.27	0.14 to 0.40	0.14 to 0.40	.00	8.04	0.0
Strong	5	0.07	–0.30 to 0.44	–0.54 to 0.68	.02	5.17	22.6
Collegiate	11	0.05	–0.29 to 0.38	1.01 to 1.11	.20	51.54	80.6

Abbreviations: I^2, Higgins and Thompson *I* statistic; *k,* number of study effects; PI, prediction interval; *Q*, Cochran *Q*; SMD, standardized mean difference; τ^2, tau squared.

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demonstrate limited agreement and proportionality between the isometric and dynamic strength change scores. Consequently, practitioners should be wary of what measure of strength is implemented following resistance training interventions, as this may alter the interpretation of the training intervention's effectiveness. Furthermore, due to the substantial variance and lack of agreement in change between these tests following training interventions, practitioners must be careful not to conflate changes in isometric strength with dynamic strength, or vice versa.

Effect of Resistance Training on Isometric Versus Dynamic Strength

While there is a considerable body of literature demonstrating isolated relationships between isometric and dynamic strength,[20,45] the difference in the *change* between forms of strength following training has received less attention.[46,47] The findings of the overall meta-analysis demonstrate that the difference between isometric and dynamic strength changes should be expected to vary dramatically. The resulting heterogeneity contrasts the generally moderate to strong cross-sectional relationships between the 2 forms of strength and indicates why it is important that cross-sectional findings are not assumed to hold true when examined across time. Indeed, the issue of transfer between isometric and dynamic strength was noted in a review by Wilson and Murphy,[15] who concluded that the vast weight of evidence suggested that changes in dynamic strength performance did not align with changes in maximal isometric strength. From an applied perspective, the findings of this present analysis provide strong evidence that it is not possible to accurately estimate a change in 1RM back squat, power clean, or deadlift from a change in isometric squat or IMTP maximal strength following future training. Furthermore, both isometric and dynamic assessments should be considered if a holistic strength diagnosis of the individual is required. Alternatively, a needs analysis can be employed to decide which strength quality is of greatest relevance and therefore prioritized in a strength assessment battery. However, other factors should also be considered when selecting appropriate strength tests. For example, rapid force production (eg, rate of force development, colloquially referred to as "explosive strength") can also be evaluated via the same isometric tests,[25,48] which may be more important than maximal isometric strength in identifying training priorities.[49-51] In addition, 1RM testing also permits the objective selection of loads for subsequent phases of training.

Only 2 interventions in a single study[38] possessed a trivial difference alongside a CI that did not envelop ±0.30. Two comparisons exhibited a significantly greater change in isometric versus dynamic strength,[41,42] while a significantly greater change in dynamic strength occurred in 5 comparisons,[36,40,43] which suggests that the isometric tests are likely to be less responsive to dynamic resistance training. It should be acknowledged that the significantly greater improvements in isometric strength in the aforementioned studies may have been influenced by the limited use of full-range lower body exercises in the final block of training. Indeed, within their respective final training period, 5 of 6 lower body lifts in one study[41] and all 7 lower body lifts in the other investigation[42] utilized a reduced range of motion at the hip and knee (eg, from the hang, ¼ squats). As such, these actions may have achieved greater specificity to the IMTP test when compared with lifts performed through larger ranges of motion. It is also possible that greater changes in isometric strength could be due to the lack of familiarization leading up to baseline testing. Further exploration is warranted to better understand the sensitivity of the IMTP and isometric squat to different forms of dynamic resistance training.

Agreement Between Isometric and Dynamic Strength Changes Following Resistance Training

The second aim of this investigation was to determine the longitudinal association between isometric and dynamic assessments of maximal lower body strength. The agreement metrics revealed very poor levels of agreement (CCC = .16) and high levels of variation (CV% = 109.27), between changes in isometric versus dynamic strength following resistance training. Based on these findings, practitioners and researchers should not use isometric and dynamic strength assessments interchangeably or infer that changes in the 2 forms of assessment are proportional. Although individual level data were obtained from only 4 studies, this resulted in pairs of change scores from 94 subjects. These data provide greater insight into the interrelationship between strength qualities than relationships assessed at a single time point.[46,47] The results of this present investigation agree with previous reports of considerably weaker relationships between different forms of strength and power-oriented tests when tracked longitudinally rather than cross-sectionally.[46,47] However, the notable finding of this investigation is that both tests are considered forms of maximal strength, yet no predictable longitudinal relationship exists between the 2 qualities in response to resistance training.

There are several differences in the underpinning mechanisms that drive force production in isometric versus dynamic actions that may explain the limited agreement revealed by the Bland–Altman analysis. Muscle activation strategies vary between isometric and dynamic actions,[52] while intramuscular and intermuscular motor unit recruitment change based on joint angle and the direction of force application.[53–56] In addition, dynamic actions are often influenced by the stretch-shortening cycle; however, this mechanism is absent from isometric tasks.[57,58] It has been suggested that musculoskeletal stiffness also has a greater contribution to force production within isometric tasks[59] and may in part explain differences between the 2 types of strength. Regardless of the mechanisms responsible, the findings of this present study provide strong evidence that these 2 forms of maximal strength represent separate neuromuscular domains. It is important to note that the vast majority of the subjects in this analysis (n = 77) undertook combined heavy-high velocity training. Further research is needed to understand how the agreement between isometric and dynamic strength change scores is impacted by different training interventions and across training phases.

Effect of Resistance Training Type on Isometric Versus Dynamic Strength

The type of training appears to have some influence on the magnitude of changes between isometric and dynamic strength. Tests of maximal dynamic strength were more sensitive to heavy strength training interventions with 7 of the 9 comparisons demonstrating at least a "small" effect or greater in its favor, while the remaining 2 comparisons were trivial. However, a considerable amount of within-comparison variation was present. The interventions in these 7 studies included a greater volume of resistance training that was similar to the dynamic strength test, which may have contributed to the outcome by way of specificity.[60,61] It is worth noting that the back squat was both a training task throughout

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the intervention and a dynamic strength assessment in each of the heavy resistance training comparisons, which likely facilitated a greater transfer effect compared with the isometric assessment.[62]

Test Type

When compared with the IMTP, the relatively narrow PI in the isometric squat subgroup indicates a more predictable relationship with maximal dynamic strength assessment. Practitioners can; therefore, be somewhat confident that dynamic strength testing is consistently more sensitive to training than the isometric squat to a small extent. The markedly wide PI in the IMTP subgroup analysis demonstrates that considerable variation in the change score differences between IMTP performance and measures of dynamic strength can be expected in future settings. Despite similar lower limb positions, there are notable differences between the IMTP and the isometric squat. A feature of the isometric squat that may not be present to the same extent in the IMTP is the high axial compressive forces that likely require different contributions from the trunk musculature.[63] In addition, unlike the isometric squat, the IMTP may incorporate a meaningful shoulder extension moment. These factors contribute to a cross-sectional unexplained variance of 43% between the 2 tests and differences in peak force ranging from 9.5% to 28.5%.[48,64] One possible explanation for the results is that the studies incorporating the isometric squat included its dynamic equivalent (ie, 1RM back squat) as the strength test more often (12/14 comparisons) than the IMTP was included alongside its dynamic equivalent (1RM power clean or deadlift) as the strength assessment (9/15 comparisons). The increased specificity between isometric and dynamic tests in the isometric squat subgroup might have enabled a more consistent association to be revealed.

Intervention Duration

Only 3 studies,[35,37,44] representing a total of 8 comparisons, included training interventions less than 10 weeks. The narrow PI and trivial effect (albeit significant) would suggest that there is little collective difference between the changes in isometric and dynamic strength across those investigations. This finding is in alignment with strong cross-sectional relationships between the 2 qualities in many cases.[20] However, notable within-comparison variability is present which indicates that the same training stimulus has markedly different effects on isometric versus dynamic strength between individuals. While the SMD was similar regardless of whether training duration was <10 or ≥10 weeks, a much greater range of effects are expected following interventions lasting 10 weeks or beyond. It is possible that longer duration training studies provide a greater opportunity for differential adaptations to manifest between forms of maximal force expression.[32,65] In other words, early-stage changes in isometric and dynamic strength are similar; however, as training time goes on, it is not possible to infer a change in one from the change in the other. Another explanation may be related to the similarity between the training interventions and both forms of assessment in the short-term studies. This included powerlifters undertaking training for competition (where success is in part determined by squat strength)[44] and interventions characterized by the squat or half squat.[37] These findings indicate that isometric strength changes may be a suitable indicator of dynamic strength changes in the short term where the training lifts and loading are highly specific to both types of assessment. However, 2[35,37] of the 3 studies did not describe the training interventions in full, making it challenging to identify the cause of the observed effect.

Population

While considerable within-study variation existed, the dynamic strength assessment was more sensitive to training than isometric tests within recreationally trained individuals when studies were pooled, albeit to a small but consistent extent. Less trained individuals have less exposure to dynamic strength activities and, therefore, likely possess a greater capacity for dynamic strength adaptation than those who are more trained.[65,66] Furthermore, in the early stages of training, more general adaptations can be expected[67,68] leading to a homogenous response across studies within this population, regardless of training. A range of potential true differences between isometric and dynamic strength changes was revealed in strong populations and may indicate that the type of strength response may be specific to the within-study characteristics (eg, the resistance training intervention) and not generalizable across situations. However, as only 2 studies containing participants whose 1RM back squat was >1.90 × body mass,[32,44] it is challenging to draw meaningful conclusions, and future investigations with strong individuals are therefore recommended.

The wide PI for the collegiate athletes subanalysis demonstrates that practitioners working with this population would be uncertain of what differences would be present in isometric versus dynamic strength changes. A range of factors such as long-term training history,[69] sport,[70] and the presence of other training tasks[71] potentially contributed to this divergent response. What is clear, however, is that changes in isometric strength do not consistently represent changes in dynamic strength, particularly within collegiate populations.

Practical Applications

Both isometric and 1RM dynamic tests are typically considered as tests of maximal strength; however, because there is substantial variance in the response between the 2 forms of strength following resistance training, and no proportional relationship, the assessments cannot be used interchangeably or assumed to assess the same quality. While practitioners may be tempted to include isometric strength measures (eg, the IMTP) in lieu of a dynamic strength test, such a decision would likely provide misleading information if dynamic strength was the attribute of interest. Consequently, when conducting a needs analysis to determine the relevance of specific physical qualities to a given sport, practitioners may wish to include both forms of assessment in their initial analysis before then considering which is of greatest relevance. As the 2 assessments represent separate strength domains, it may also be necessary for practitioners to use specific training interventions depending on whether isometric or dynamic strength is targeted.

Conclusions

The results of this systematic review and meta-analysis demonstrate that a range of differences between changes in maximal isometric strength (assessed via the isometric midthigh pull and isometric squat) versus maximal dynamic strength (assessed via the 1-repetition-maximum back squat, deadlift, and power clean) can be expected following resistance training. Additionally, intervention duration and specificity of the training tasks to the test type, particularly in the final phase of training, appear to have some

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impact on the observed effect. A key outcome of this investigation was the absence of agreement and proportionality between changes in isometric and dynamic strength, which provides strong evidence that these 2 forms of strength represent separate neuromuscular domains.

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